

MTE 203 – Advanced Calculus

Week 4 Tutorial Solutions

Problem 1: Midterm Exam Problem 4 Fall 2016

A bead of mass " m " slides without friction down a wire bent in the shape of the curve $y = x^2$ under the influence of the gravitational force $-mg\hat{j}$.

The speed of the bead as it passes through the point $(1,1)$ is " V ". At that instant, find

- The normal acceleration of the bead in terms of " V "
- The rate of change of its speed in terms of " g ".

Solution: Please see pdf accompanying this file

Problem 2: [S.12.1, Prob. 27]

- A company wishes to construct a storage tank in the form of a rectangular box. If material for sides and top costs $\$1.25/\text{m}^2$ and material for the bottom costs $\$4.75/\text{m}^2$, find the cost of building the tank as a function of its length l , width w and height h .
- If the tank must hold 1000 m^3 , find the construction cost in terms of l and w .

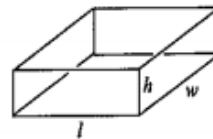
Solution

(a) The cost in cents is

$$C = 125(2wh + 2lh + lw) + 475lw = 600lw + 250h(l + w).$$

(b) If $V = 1000 = lwh$, then $h = 1000/(lw)$, and

$$\begin{aligned} C &= 600lw + 250 \left(\frac{1000}{lw} \right) (l + w) \\ &= 600lw + \frac{250\,000(l + w)}{lw}. \end{aligned}$$



Problem 3 [S.12.1, Prob. 17]

Draw the level curves of $f(x, y) = y - x^2$ in a contour map corresponding to the values $f(x, y) = -2, -1, 0, 1, 2$

Solution:

The domain of the function will be the entire x-y plane. From observing the function, it can also be seen that there's no constraints on its range either. Rearranging the equation as

$$f(x, y) = z = y - x^2$$

Solving for the different values of $f(x, y)$ and rearranging them,

$$f(x, y) = -2 = y - x^2 \rightarrow y = x^2 - 2$$

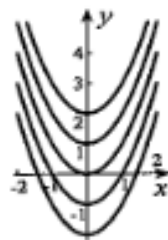
$$f(x, y) = -1 = y - x^2 \rightarrow y = x^2 - 1$$

$$f(x, y) = 0 = y - x^2 \rightarrow y = x^2$$

$$f(x, y) = 1 = y - x^2 \rightarrow y = x^2 + 1$$

$$f(x, y) = 2 = y - x^2 \rightarrow y = x^2 + 2$$

Drawing these functions in the x-y plane, we can see that the contour curves are formed by shifting parabolas:



The 3D surface and the contour map generated using matlab is shown below.

